

# Where a Kalshi edge could actually persist

STRATEGY RESEARCH | 24 SEPTEMBER 2026 | PREPARED FOR LOGAN

**A mechanism can be real while a particular bot is unprofitable. None of these ten has demonstrated live, net profitability for this account.**

## The conclusion

The strongest direction is a small portfolio of specialized trading processes, each exploiting an identifiable source of mispricing or compensation. The research does not establish a deployable cash cow. It does identify ten falsifiable candidates, three near-term priorities, and one unusually interesting new opportunity in perpetual futures.

## What changed after this review

Sethi's work supports evaluating forecasts by the trading decisions they induce, with explicit risk preferences. It does not imply that a more accurate model necessarily earns more money. The best fit is a specialist that combines an independent forecast with market information and abstains unless the disagreement survives costs. Less prominent markets deserve more attention than the headline contracts everyone studies.

## The immediate allocation

Concentrate research on (1) neglected-market hybrid forecasts, (2) station-specific weather nowcasts, and (3) selective liquidity-reward allocation. Give perpetuals a short, separate feasibility screen for account economics and executable latency. Do not spend heavily on speed infrastructure before that screen passes. Preserve the NFL engine as a benchmark while resolving the Q7 implementation conflict.

## What was reviewed

The 74-page Engines of Prophecy deck; the Grokbot operations repository and its link to the canonical GPT-6-Astra science repository; main plus four relevant research branches; the named authors' relevant primary-source work; and current Kalshi documentation, incentives, fees, RFQs and settlement mechanics. This is an evidence and strategy audit, not a rerun of every experiment or an audit of a running production host.

## How to read the ranking

The ranking is a judgment about the value of the next research dollar, combining evidence, testability, repetition, competition and fit with the existing work. It is not an expected-return forecast. Each card distinguishes sourced findings from my proposed adaptation, identifies who could pay for the edge, and gives an experiment that can reject it.

# The ten candidates

## RANKED BY RESEARCH PRIORITY, NOT PROMISED RETURNS

| Rank / candidate                     | Potential source of profit                             | Evidence / principal obstacle                                          |
|--------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------|
| 1. Neglected-market hybrid forecasts | Better processing of specialist information            | Historical forecast-profit evidence; no current executable replication |
| 2. Station-specific weather nowcasts | Better conditional distribution and settlement mapping | Official measurement mechanics; forecast edge untested                 |
| 3. Selective liquidity rewards       | Exchange-funded resting liquidity                      | Current program + measured gaps; competition and losing fills          |
| 4. Subsidized-perps stale quotes     | Faster takers against committed makers                 | Fresh mechanism evidence; fees, access and latency unknown             |
| 5. Calibrated anti-longshot making   | Demand for unlikely, exciting outcomes                 | Historical Kalshi evidence; selection and fee-regime changes           |
| 6. Official-source interpretation    | Faster understanding of precisely defined outcomes     | Plausible information advantage; open-market windows scarce            |
| 7. Correlation-aware combo quoting   | Demand for packaged outcomes                           | Native RFQ mechanism; joint-model and inventory risk                   |
| 8. Anchored flow dislocations        | Temporary pressure from directional demand             | Historical trader ecology; distinguishing information from noise       |
| 9. Crypto settlement-window pricing  | Correct use of already-fixed average observations      | Current benchmark feed; competition and tiny cost budget               |
| 10. Rule-proven event baskets        | Cross-contract inconsistency                           | Mathematical payoff bounds; static repo scan found no gross edge       |

### Evidence labels used on the cards

Historical empirical evidence means an author studied past data; it is not a current replication. Mechanism verified means official rules or market design support the mechanism. Repo measured means the saved study contains observations, subject to its stated sample and execution assumptions. Proposed means the trading adaptation is my inference. None is labeled profit-validated.

# What the repositories actually establish

## READ RESULTS AT THEIR RECORDED COMMIT AND IMPLEMENTATION

**Preserve every frozen negative. A new market, modified queue or renamed arm does not create a fresh holdout.**

### Repository roles

Grokbot-Deathmatch-Dedicated-Repo is an operations and handoff repository, not a second independent strategy implementation. It names GPT-6-Astra-Deathmatch as its science source of truth. I did not find a distinct third science repository in those instructions. Operations notes describe a GET-only recorder; that is not verification that a remote process remains healthy now. [R1, R2]

### Execution dominates the old headline

The saved NFL progression goes from a legacy +\$1,027.47 replay to a repaired +\$681.73, then approximately +\$44.55 when the queue assumption is increased from 290.595 to 3,300. That is roughly a 93.5% reduction from the repaired result, on a reused 16-game week. It demonstrates fragility to execution assumptions, not a repeatable account return. The later 31-game factorial still reuses development data. [R2]

### Q7 is not one consistent result

Main retains a frozen/not-run Q7 study and later refiner diagnoses reporting B/D retention around 0.843. The paired-price research branch reports its router-on arm ahead of router-off under its replay stresses. The policies differ: main gates quote creation around existing working orders; the research branch filters inside selection and applies bounded sizing. Reconcile code, fixture, arm labels and full ledgers before comparing them. The discrepancy cannot be solved by selecting the better P&L. [R2, R3]

### Negative transfer is useful evidence

Q8 joint routing improves the lighter-queue replay but loses its advantage under higher queue; the preregistered promotion screen fails. M2's four CFB and four WNBA games do not establish successful NFL transfer. M3's better CFB timing is four reused games; a changed queue assumption is not an algorithmic improvement. Many newer market-family and microstructure labs contain unit tests, null scorecards or synthetic fixtures rather than accepted market evidence. [R4, R5]

### The widest scan narrows the opportunity

AMS tested 216 static nested-threshold cases, of which 180 had complete depth: zero positive gross payoff-floor opportunities. Its completed incentive survey had 199 usable open markets from a prespecified 200-market sample; 32 lacked target depth on at least one side. A later filter left only three with ideal remaining half-pool rewards above one \$10 losing fill. Those are candidate observations, not net profits or independent validation. [R6]

# What the deck and authors contribute

## FULL DECK REVIEW; PRIMARY-SOURCE CORPUS WITH ACCESS LIMITS

**The most actionable Sethi lesson: compare complete trading policies at available prices, with risk preferences and costs, rather than selecting the best forecaster in isolation.**

### Pages 1-32: prices are not pure probabilities

The deck moves from market design to the interpretation of prices and forecast scoring. Prices reflect beliefs, budgets, risk preferences and participation. Its Brier-score examples motivate comparing mechanisms, but a score alone does not tell us which executable trades are attractive. Reciprocal YES and NO quotations are two views of one book, not independent arbitrage legs. [D1]

### Pages 33-44: profitability and trader specialization

The virtual-portfolio approach converts forecasts into risk-sensitive trades. The election results differ across market types: gains in some less visible contests coexist with losses in headline contests. The historical Intrade examples show that trading style and capital use matter; maximum posted margin is not initial capital and should not be converted into a sensational ROI. These are simulated or historical findings, not evidence for today's Kalshi execution. [D1, S01-S04]

### Pages 45-74: reflexivity, volume and settlement

Large trades need not be uninformed; manipulation may even influence the event being predicted. Wash-trading analysis requires a mechanism and counterparty evidence, not merely repeated trade sizes. Late-2020 election prices illustrate persistent disagreement and limits to arbitrage, including settlement uncertainty and tied-up capital. The closing material reinforces conditional judgment rather than a universal bot recipe. [D1, S05-S07]

### Named-author coverage

Rajiv Sethi, Julie Seager and Emily Cai connect the 2021 model-market precursor to the 2024 profitability paper. Daniel M. Benjamin and Fred Morstatter also contribute relevant human-machine forecasting research. Olivia Bobrownicki, Yuqi Cheng, Anushka Kumar and Anusha Wanganoo appear on the 2024 paper. Anna Hammell, Tianshuo Liu, Sachi Patel and Ramya Subramanian appear on the 2025 election paper. All 13 names were included in targeted searches; no additional independently verified trading corpus was established for every junior coauthor.

### What I could and could not read

I inspected all 74 slides and read the available full 2021 precursor and SAGE study, plus accessible primary abstracts, author commentary and current exchange documentation. The 2024 and 2025 ACM/SSRN full texts were not retrievable through the available paths; their primary university abstracts and the deck's displayed results were accessible. The report does not present abstract-only access as full-paper review or claim to exhaust every author's publications.

# 01 Neglected-market hybrid forecasts

**FIRST PRIORITY | HISTORICAL EMPIRICAL SUPPORT; CURRENT EDGE UNTESTED**

**Best strategic fit: a narrow specialist with permission to abstain, not a general-purpose model that trades every disagreement.**

## **The edge and its payer**

Specialize in markets where an independent, disciplined forecast captures information that the marginal trader has not fully processed. Combine it with market prices instead of mechanically opposing them. The proposed source of profit is uneven research attention, not the belief that markets are generally foolish.

## **What supports it**

The 2025 Sethi-coauthor study reports models losing in prominent presidential markets while some outperform in less visible markets; the uploaded deck displays that contrast. The 2021 precursor finds benefits from combining models and markets. SAGE supports skill- and recency-sensitive human-machine aggregation, but tests forecasting scores rather than trading profit. These are reasons to investigate selective hybrids, not a guarantee that low attention implies inefficiency. [S01-S03, D1]

## **Adaptation for this project**

Choose one coherent family with defensible external information: initially selected down-ballot elections, or another family only after verifying comparable forecast inputs. Create a timestamped probability distribution with market-only, model-only and hybrid baselines. Weight forecasts using past out-of-sample calibration; shrink weak evidence toward the market. An LLM can extract and organize public information, but its unsupported confidence is not a probability model.

## **Economics and capacity**

Trade only when conservative expected payout exceeds the executable cost plus all fees, a model-error allowance and a capital charge. A 54% estimate against a 50c ask gives only 2.25c before other costs under an illustrative standard 1.75c taker fee. Election capital can remain locked for weeks; apparently large percentage gaps may generate little annual dollar capacity.

## **First validation and rejection rule**

Freeze one family and its forecast sources before observing future outcomes. Record revisions and orderbooks at actual receipt time. Compare the three baselines under identical feasible fills, then test the hybrid on untouched events. Reject if its gains disappear against market-only, are concentrated in one correlated event, or exist only at midpoints. Treat this as a slower research stream; one election cycle is not hundreds of independent experiments.

# 02 Station-specific weather nowcasts

**HIGH PRIORITY | SETTLEMENT MECHANICS VERIFIED; FORECAST ADVANTAGE PROPOSED**

**The defensible asset would be a clean station-to-settlement dataset and calibrated residual model, not access to a common weather API.**

## **The edge and its payer**

Estimate the distribution of the final official daily high or low at the exact settlement station, conditional on the weather already observed and what can still happen. The potential counterparty error is using a generic city forecast, a rounded app observation or an unconditional morning distribution after the day has substantially unfolded.

## **What supports it**

Kalshi's weather guide distinguishes final NWS daily climate reports from hourly products and explains the local-standard-time day. NWS explains how five-minute averages and Celsius conversion can make displayed temperatures differ from official climatic extremes. These establish a real modeling problem; they do not establish that current prices systematically mishandle it. The repo's weather scaffolds do not yet provide a profitable out-of-sample execution study. [S08, S09, R2]

## **Adaptation for this project**

Start with a few stations rather than every weather market. Model the remaining-day maximum conditional on the observed maximum, time, clouds, wind shifts, humidity and forecast changes. Map raw observations into the official measurement and rounding process. Learn station- and regime-specific residuals. Treat missing or stale feeds as missing evidence; the NWS Phoenix experimental page itself currently warns about outages and erroneous preliminary values.

## **Economics and capacity**

Prefer situations with a meaningful probability gap, not a one-tick race. There is daily repetition, but neighboring strikes and cities can share weather risk. Prices in apparently impossible buckets are not free money when the observation feed differs from the settlement source. Account for settlement lag and the chance of losing all principal on an incorrect near-certainty trade.

## **First validation and rejection rule**

Archive original forecasts, receipt-time observations, exact station rules, books and eventual final reports for an initial 60-day pilot, then reserve future days for evaluation. Compare against the market and a simple forecast-error distribution. Sixty days is a collection target, not proof of power. Reject if extra skill fails to improve executable net returns, rounding explains the apparent edge, or profits depend on one season or station.

# 03 Selective liquidity-reward allocation

**HIGH PRIORITY | PROGRAM VERIFIED; REPO GAP OBSERVATIONS; NET EDGE UNTESTED**

**Potential small, repeatable supplement; currently weak evidence for a broad cash cow. Do not annualize an ideal hourly rate from one snapshot.**

## **The edge and its payer**

Allocate resting liquidity only where the expected exchange reward exceeds adverse selection, inventory loss, fees and capital use. Kalshi funds the subsidy. This is a scheduling and competition problem across eligible markets, not a forecast-free passive-income strategy.

## **What supports it**

The current program scores qualifying depth on both sides and pays proportional shares over eligible snapshots. The repo's AMS004 survey found 32 of 199 usable markets short of target depth. Of 29 later-screened one-empty-side cases, 16 would make a 1c order cross; only 13 could rest. Three passed an idealized remaining-reward-versus-\$10-loss screen. This validates sparse candidate supply, not a durable income stream. [S10, R6]

## **Adaptation for this project**

Continue the existing census/tape work instead of launching a duplicate snapshot study. Recompute the reference level and reward share after adding our proposed order, and track competitors, program windows, eligibility and minimum payment. Rank candidates by net reward per constrained capital-hour. Inventory acquired by a fill changes the decision: the bot must not replenish losing exposure indefinitely to preserve reward share.

## **Economics and capacity**

If one trader owns all qualifying score on one side, its ideal share is half the pool over the relevant time, not the whole pool. A 1c bid for 1,000 contracts risks \$10 principal on a full losing fill, plus costs. The three saved candidates' ideal gross rates were only about \$0.32-\$0.65 per hour. Repeated losing fills, competition or short windows can dominate that.

## **First validation and rejection rule**

Measure duration of reward gaps with a forward tape, not just frequency in a snapshot. Simulate orders entering the score denominator and carry inventory honestly. Keep a no-reward P&L control and a realized-reward ledger. Reject markets whose conservative reward is below expected inventory loss, whose qualification depends on unverifiable assumptions, or whose gaps disappear before an order can rest. Small pilot orders would require a separate trading decision after research approval.

# 04 Stale quotes in subsidized perpetuals

## HIGHEST CONDITIONAL CEILING | NEW MECHANISM EVIDENCE; ACCESS AND SPEED UNPROVEN

**Worth an immediate feasibility screen. Its potential is large only conditional on a demonstrable speed advantage and viable account economics.**

### The edge and its payer

A faster trader identifies when a committed market maker's resting perpetual-futures quote is stale relative to executable fair value. A liquidity subsidy can keep the maker quoting despite selection losses. Our proposed trade monetizes that selection loss while controlling hedge, basis and exit costs.

### What supports it

Sethi's September 23 commentary discusses volume generation through this mechanism. Kalshi's September 22 explanation says the questioned trades involved profitable takers against a maker and describes liquidity obligations plus a temporary fee holiday for qualifying perpetuals self-clearing members. This is useful first-party mechanism evidence, not an independent audit of participant returns. The article explicitly says a circulating +0.3/-0.3 basis-point fee program is not live. [S11, S12]

### Adaptation for this project

Separate this from binary BTC betting. First verify the actual account's fee tier, any self-clearing requirements, capital contribution and accessible venues. Record synchronized external executable bids/asks, Kalshi perps books and source/receipt timestamps. Test whether a signal remains actionable after our realistic order-arrival delay, not merely whether one feed leads another.

### Economics and capacity

Perps fees apply to notional exposure; opening, closing, hedging, funding and basis matter. A fee holiday available to an institution is not automatically available to this account. Margin leverage does not increase the underlying edge and can magnify losses. The repository's external spot-data access failure blocks a local feasibility claim; it does not disprove the mechanism. [S13, R6]

### First validation and rejection rule

Run a brief read-only feasibility study before infrastructure spending: obtain valid feeds, establish costs, and count stale quotes still present at conservative arrival times. Report attainable depth and hedge slippage. Reject if the apparent spread disappears at arrival, fees consume it, or profitable execution requires inaccessible membership economics. Any later execution study must distinguish failed orders from assumed fills.

# 05 Calibrated anti-longshot market making

## EMPIRICAL PRIOR | HISTORICAL KALSHI PATTERN; CURRENT REPLICATION REQUIRED

**A defensible inventory style if replicated. The historical result does not justify a universal "buy 90c favorites" bot.**

### The edge and its payer

Supply liquidity to demand for low-probability, exciting outcomes only when their price exceeds a conservative calibrated probability. Buying NO is one way to express the view; blindly buying every favorite is not the strategy. Potential compensation comes from lottery-like demand and liquidity provision.

### What supports it

Burgi, Deng and Whelan's January 2026 paper studies Kalshi data through April 2025 and documents favorite-longshot bias and better maker than taker outcomes. Its historical fee regime and its restriction to markets open at least 24 hours limit transfer to today's fast markets. The evidence is an empirical prior, not an executable maker backtest for our queue or a guarantee that the bias remains. [S14]

### Adaptation for this project

Estimate calibration by family, horizon, price, liquidity and maker/taker side. Use the existing FL harness only after loading accepted real data and actual fee versions. Require an independent probability check and adverse-selection model. Avoid treating 20 correlated longshots on one underlying event as 20 diversified positions.

### Economics and capacity

A hypothetical NO at 95c with a true 98% win probability has 3c gross edge. At the illustrative standard taker multiplier, its unrounded fee is 0.3325c, leaving 2.6675c before other costs. If its true probability is only 95%, the same trade loses after fees. A high win rate can hide occasional losses large enough to erase many wins.

### First validation and rejection rule

Preregister price bands and market families, then evaluate future settlements and fill-conditional returns. Compare equal-risk maker, taker and abstention policies with the same underlying signals. Attribute return separately to calibration, spread and rewards. Reject if the estimated bias is eliminated by current fees, selected fills lose more than unselected quotes, or tail concentration dominates the return.

# 06 Official-source interpretation

## PROMISING INFORMATION HYPOTHESIS | AVAILABILITY MUST BE VERIFIED PER CONTRACT

**The initial product is an eligibility-and-interpretation test. If markets close before the information arrives, stop that family immediately.**

### The edge and its payer

Translate an official release into the exact contract outcome more reliably and quickly than traders reacting to a headline. Candidate difficulties include fiscal versus calendar periods, reported versus adjusted earnings, growth-rate definitions, rounding and the contract's source hierarchy. The proposed payer is a trader who interprets the wrong number or remains uncertain after the correct one is available.

### What supports it

The project's all-market triage retained company-source interpretation while finding that its tested CPI contract closes before the release. That blocks the simple post-release CPI-sniping story. SAGE supports disciplined evidence aggregation, but supplies no proof of a latency-trading edge. This card is a proposed adaptation with a clear gate: the relevant information must arrive while the specific market can still execute. [R6, S03]

### Adaptation for this project

Build a small registry of contract definitions and approved public source URLs. Before each event, identify the exact field or sentence needed. Archive the first received version of the release and deterministic extraction output. Use an LLM for ambiguous-text triage, with explicit abstention; use numerical parsers and rule checks wherever possible. Never treat a revised web page as if it was available earlier.

### Economics and capacity

The opportunity may last seconds or minutes, but event frequency and available depth may be small. The best niche could be semantic difficulty rather than raw network speed. A price approaching certainty still carries source, interpretation and settlement risk. Fees and failed orders belong in the measured result.

### First validation and rejection rule

Screen the first 20 candidate releases for actual open-market overlap before building a strategy. For those that pass, record source arrival, interpretation completion, book updates and attainable prices; compare a simple headline parser with the rule-aware parser. Reject any family with no tradable information window, no incremental semantic advantage, or unresolved rule ambiguity. Move to untouched future releases before claiming returns.

# 07 Correlation-aware combo quoting

## NATIVE MECHANISM VERIFIED | JOINT-PROBABILITY AND EXECUTION EDGE PROPOSED

**A potentially scalable specialist business, but a later build than weather or reward measurement because model and inventory demands are materially higher.**

### The edge and its payer

Quote packages of outcomes when a calibrated joint model can price the requested combination more accurately than competitors while charging enough for inventory and selection risk. A plausible payer is a customer seeking a convenient or appealing package. The edge would be joint pricing and risk management, not multiplication of marginal probabilities.

### What supports it

Kalshi supports dedicated combo markets and RFQs. Its documentation describes private maker responses and shorter confirmation/execution windows for combos. The repo already recognizes combo-versus-leg-product measurement, but does not establish a profitable dealer. Official rules validate an execution route; they do not show that existing dealers misprice correlation. [S15, S16, R2]

### Adaptation for this project

Begin with one sport and a limited set of combinations whose dependence can be estimated. Simulate common underlying game states so that leg probabilities and joint probabilities are internally consistent. Track pushes, nonparticipation and fractional leg payoffs exactly. Quote only packages for which the full requested size fits portfolio limits; RFQ acceptance is not a guaranteed neutral hedge.

### Economics and capacity

If both legs have probability 0.60, independence gives 0.36, but a plausible positively dependent joint probability might be 0.45. Selling the package at 0.40 then has negative expected value despite looking expensive relative to 0.36. Marginal legs generally do not replicate a conjunction exactly. Fees, confirmation timing, leg liquidity and common-factor exposure can overwhelm apparent pricing differences.

### First validation and rejection rule

Collect accessible RFQ opportunities and contemporaneous legs under actual account permissions. Fit the joint model before evaluation, then compare it with independence and a conservative dependence bound. Test accepted-quote selection rather than all displayed opportunities. Reject if correlation uncertainty spans the quote margin, executable volume is inaccessible, or residual unhedgeable inventory supplies all apparent profit.

# 08 Anchored flow-dislocation trading

## HISTORICAL MICROSTRUCTURE SUPPORT | SIGNAL IDENTIFICATION UNPROVEN

**Useful only with a credible anchor. "A whale moved the market" is an observation, not a trading thesis.**

### The edge and its payer

Trade temporary price pressure from urgent or strongly directional demand only when an independent value estimate and related markets remain stable. The proposed profit source is compensation for absorbing inventory until pressure recedes. This is a conditional mean-reversion hypothesis, not a claim that large traders are uninformed.

### What supports it

Rothschild and Sethi's Intrade study describes heterogeneous strategies, including market making, arbitrage and directional activity. Sethi's commentary emphasizes that a large order can reveal information or reflect belief, and the deck discusses reflexive effects. Those sources motivate distinguishing trader pressure from news; they do not provide a ready-made whale-fading signal. [S04, S05, D1]

### Adaptation for this project

Use public book and trade information to detect abrupt imbalance, consumed depth and divergence from a preregistered anchor. Do not invent private trader identities from anonymous Kalshi prints. Require source monitoring and coherent comparison contracts. Let the independent model veto reversion trades when news explains the move; add explicit time and inventory limits.

### Economics and capacity

Spreads and short-horizon reversals can look attractive before fills. A maker is most likely to fill when its quote has become bad, so quote-level reversion is not the relevant return. Measure conditional adverse movement after attainable fills. Position capacity is bounded by both event risk and realistic exit depth; a marked gain in an illiquid book may be uncollectible.

### First validation and rejection rule

Preregister a small number of imbalance/anchor conditions and compare with matched no-shock periods. Use future events, conservative queue assumptions and news receipt times. Test whether reversion exceeds costs after actionable arrival, not at the detected extreme. Reject if effects vanish when news is controlled, only one event drives the result, or forced exits produce negative net returns.

# 09 Crypto settlement-window pricing

## BENCHMARK DATA MECHANICS VERIFIED | REMAINING-UNCERTAINTY EDGE PROPOSED

**A sharply defined falsifiable variant worth testing; no basis yet to resurrect a generic short-term crypto bot.**

### The edge and its payer

Price a short-horizon binary using the exact settlement average and the part of that average already observed. The target is a trader or model treating the contract as a bet on a final spot tick or using the wrong reference feed. This differs from a generic BTC momentum or indicator bot.

### What supports it

Kalshi now documents an authenticated CF Benchmarks feed with raw index values, trailing averages and final-minute quarter-hour accumulation. The once-per-second documentation specifies boundary conventions; the five-Hz channel has raw ticks rather than those averages. That supports a rule-faithful data pipeline. It does not establish that the market ignores the same public information. Contract-specific rules still govern settlement. [S17, S18]

### Adaptation for this project

For a contract whose rule is an equal-weight average of 60 specified observations, track the fixed sum  $S$  after  $k$  observations. To finish above strike  $K$ , the remaining observations must sum to more than  $60K - S$ , subject to the contract's tie and rounding rules. Model that remaining joint path distribution; do not count correlated seconds as independent draws or replace it with the final tick.

### Economics and capacity

Most of the uncertainty can disappear near close, but so can accessible price discrepancies. Many participants have the same feed, fees consume small margins, and missing ticks can create false near-certainties. Trading and settlement timestamps must be distinguished. A public benchmark stream alone is not a speed advantage.

### First validation and rejection rule

Reconstruct archived outcomes from the precise source and compare against settled contracts before estimating profit. Then compare spot-only, full-window and partial-window models on future windows at realistic order arrival. Require real books and fee versions in the existing BTC harness. Reject if replay cannot reproduce settlement, incremental calibration does not survive costs, or fills are assumed after trading has closed.

# 10 Rule-proven event baskets

## MATHEMATICAL SUPPORT | NEGATIVE STATIC SCAN; CONDITIONAL BACKLOG

**Keep as a selective structural check and backlog item. The existing negative scan makes it a poor first use of additional build time.**

### The edge and its payer

Exploit a price inconsistency among contracts with a proven relationship: a threshold implies another threshold, deadlines nest, or a complete set partitions all outcomes. Potential profit comes from fragmented attention or the service of completing an inconvenient basket.

### What supports it

Payoff inequalities are exact when the rules truly imply them. But the project's AMS static scan found no positive gross floor in 180 depth-complete cases from 216 candidate cases. That is evidence against promoting the already-tested static taker scanner. Sethi's work on arbitrage limits also cautions that apparent price differences can persist because implementation is costly. [R6, S06]

### Adaptation for this project

Keep only genuinely new relationships or a distinct execution policy, such as patient basket completion during transient imbalance. Prove the relationship with identical source, timestamp, threshold convention and resolution hierarchy. Do not conflate YES/NO reciprocity with two independent legs. A paper deadline implication can fail if contracts use different definitions of occurrence or announcement.

### Economics and capacity

For A contained in B, YES(B) plus NO(A) pays at least \$1, provided their binary rules are compatible. Buying those legs for 49c and 48c appears to lock 3c before fees, but illustrative standard taker fees total about 3.4965c, making the floor negative. Maker quotes may improve arithmetic but introduce non-atomic fills, unmatched inventory and queue risk.

### First validation and rejection rule

Before collecting more data, specify what differs from the failed static scan. Test new rule classes separately from execution changes. Track matched and unmatched legs, cash usage and liquidatable value with shared depth. Reject if the floor is not positive after all costs at feasible size, if rule equivalence is uncertain, or if expected unmatched losses consume the completion margin.

# The accounting that decides survival

## ONE SHARED CAPITAL POOL; EXECUTION-CONDITIONAL RETURNS

**Optimize risk-adjusted net dollars per constrained capital-hour, with uncertainty and capacity reported beside the headline P&L.**

### Binary expected value

For a fully paid YES purchase held to a binary settlement, expected net value per contract is  $q - p - \text{net fees} - \text{other costs}$ . Here  $q$  must be the probability conditional on the information and the fact that our order fills. A maker's unconditional forecast can be right while its filled orders are systematically bad. Exit-based strategies must model realized entry and exit cash flows instead of replacing them with terminal payoff.

### Fees must be versioned

The July 7 event-contract schedule gives model formulas with market multipliers; maker fees are not universally zero. Illustrative unrounded standard-multiplier taker cost at 50c is 1.75c. Current API documentation distinguishes direct-member balance precision of \$0.0001 from non-direct precision of \$0.01 and describes per-order rounding accumulators. The repo's older cent-ceiling convention requires a version-and-account reconciliation, not an assumption that one formula covers every historical and current trade. Prefer actual recorded fee components. [S19, S20]

### Separate the profit sources

Report trading P&L excluding rewards, exchange rewards actually earned, fees, inventory liquidation or settlement, and capital cost. Do not hide a negative forecasting strategy inside a positive subsidy. Likewise, do not count an ideal reward estimate as cash received. For perpetuals use notional-based fees, funding and hedge costs, not the binary fee formula.

### Capacity is a first-class output

For each candidate record executable dollars per day, independent opportunities, capital-hours, stress drawdown and correlation with existing positions. A 20% return on \$5 of available depth is a curiosity; a modest repeatable edge with substantial capacity could be more valuable. A quote can disappear, and consuming it changes the opportunity.

### No duplicate liquidity or fake diversification

A public print cannot be credited as a fill to multiple bots, and competing bots must share cash, queue and exit depth. Contracts on one election, game, weather system or crypto price path share risk. Event-cluster uncertainty matters more than the raw number of fills or timestamps. Never annualize a reused-week replay into an account-growth claim.

# A research program that can say no

## PROPOSED EXPERIMENT ORDER; NO LIVE ORDERS PLACED

**Research should cheaply eliminate weak candidates. Building ten production bots before validating one would spread the same uncertainty across ten codebases.**

### First: repair the decision record

Pin main and research heads. Reconcile Q7's policy definitions, arm labels and scorecards without retuning the frozen studies. Add a common fee/account-version manifest. Mark each study as unit-only, synthetic, historical replay, prospective shadow or live. The existing code and negatives are assets only when these distinctions remain visible.

### Next: three focused workstreams

Use the neglected-market hybrid as the strategic forecasting stream, with explicit awareness that resolution is slow. Use weather as the daily repeated physical-data stream. Continue the existing reward census and prospective tape as the near-term microstructure stream. Give perps a limited feasibility screen in parallel only if valid data and actual fee access can be obtained. This is a proposed allocation of research effort, not a new authorization to trade.

### Minimum preregistration

Freeze the market universe, exclusions, receipt-time information set, fee regime, order timing, sizing, fill model, stopping rules and evaluation metrics before the test. Limit candidate variants; log every attempted variant. Separate discovery, tuning and untouched evaluation periods. Sampling more prints from the same event does not solve small-sample uncertainty.

### The common scorecard

Track net P&L with and without rewards; calibration; fill rate; adverse selection after fills; feasible versus requested size; unresolved inventory; capital-hours; drawdown; and event concentration. Compare with market-only, simple-model and no-trade controls where relevant. Report uncertainty by event or day and sensitivity to worse queues, latency and costs. No universal sample count substitutes for enough independent events and stable effects.

### Promotion and rejection

Promote a candidate only after an untouched prospective evaluation is economically positive under realistic costs and conservative execution, the uncertainty is acceptable for its risk, and the result survives reasonable stress without dependence on one exceptional event. If not, reject or record the precise missing evidence. Do not revise a threshold on the evaluation data and keep calling it a holdout. A later tiny live test would calibrate fills, not retroactively validate an overfit simulation.

# Sources and access notes

## PRIMARY SOURCES 1-8 OF 24 | ACCESSED 24 SEPTEMBER 2026

### S01

Sethi et al. (2025), Political Prediction and the Wisdom of Crowds. DOI 10.1145/3715928.3737483. Primary USC abstract; full-paper access unavailable. Deck supplies displayed result tables.

[Open primary source](#)

### S02

Sethi, Seager, Cai, Benjamin and Morstatter (2021), Models, Markets, and the Forecasting of Elections. Full arXiv paper examined; historical virtual portfolios and hybrid forecasts.

[Open primary source](#)

### S03

SAGE: A Hybrid Forecasting System for Predicting Geopolitical Events (AI Magazine; arXiv version 2024). Full text examined. Forecasting experiment, not a trading-profit test.

[Open primary source](#)

### S04

Rothschild and Sethi (2016), Trading Strategies and Market Microstructure: Evidence from a Prediction Market. Primary journal abstract and deck illustrations.

[Open primary source](#)

### S05

Sethi (2024), Forecasting Mechanisms: An Update. Primary author commentary on forecasts, large trades and inference.

[Open primary source](#)

### S06

Sethi, The Limits of Arbitrage. Primary author commentary; implementation limits are not proof that a discrepancy is monetizable.

[Open primary source](#)

### S07

Sirolly, Ma, Kanoria and Sethi, Network-Based Detection of Wash Trading. Columbia primary research page, supplemented by the deck. Historical Polymarket evidence does not identify current Kalshi traders.

[Open primary source](#)

### S08

Kalshi, Weather Markets. Current official settlement-source and climatic-day guidance, read September 24, 2026. Each market rule remains authoritative.

[Open primary source](#)

# Sources and access notes

## PRIMARY SOURCES 9-16 OF 24 | ACCESSED 24 SEPTEMBER 2026

### S09

NWS Phoenix, High-Resolution KPHX ASOS Data. Primary measurement/rounding explanation; current page also warns about feed availability and preliminary errors.

[Open primary source](#)

### S10

Kalshi, Liquidity Incentive Program. Current official mechanics, eligibility, score weighting and payment conditions, read September 24, 2026. Program terms may change.

[Open primary source](#)

### S11

Sethi (September 23, 2026), Volume Inflation Without Wash Trading. Fresh author interpretation of subsidized perps liquidity; not audited participant P&L.

[Open primary source](#)

### S12

Kalshi (September 22, 2026), The Facts Behind Kalshi's Perpetuals Volume. Venue explanation of maker incentives, taker selection and fee-holiday scope. Fictional program examples are not account offers.

[Open primary source](#)

### S13

Kalshi, Perps Fees Explained. Current official description of notional fees and separate funding; actual account tier still requires verification.

[Open primary source](#)

### S14

Burgi, Deng and Whelan (January 2026), Makers and Takers: The Economics of the Kalshi Prediction Market. Full 44-page author-hosted paper examined; historical period ends April 2025.

[Open primary source](#)

### S15

Kalshi, Combos. Current official market, RFQ and settlement mechanics; product of leg payoffs is not independence of leg probabilities.

[Open primary source](#)

### S16

Kalshi API, Request for Quote (RFQ). Current official interfaces, full-size quotes, confirmation and execution timing.

[Open primary source](#)

# Sources and access notes

## PRIMARY SOURCES 17-24 OF 24 | ACCESSED 24 SEPTEMBER 2026

### S17

Kalshi API, CF Benchmarks Value Feed. Current official source, timestamps, availability and average-window conventions.

[Open primary source](#)

### S18

Kalshi API, CF Benchmarks 5Hz Value Feed. Current official raw-tick channel; distinct from the per-second average fields.

[Open primary source](#)

### S19

Kalshi Fee Schedule, effective July 7, 2026. Downloaded 12-page schedule; market multipliers and fee versions matter.

[Open primary source](#)

### S20

Kalshi API, Fee Rounding. Current official member-type precision, fee components and accumulator mechanics.

[Open primary source](#)

### S21

Sethi et al. (2024), Evaluating Prediction Mechanisms: A Profitability Test. DOI 10.1145/3643562.3672612. Primary Nova abstract and deck; ACM/SSRN full text inaccessible.

[Open primary source](#)

### S22

Sethi and Vaughan, Belief Aggregation with Automated Market Makers. Primary Microsoft Research publication page. Theory concerns cost-function makers, not a ready-made Kalshi limit-order-book strategy.

[Open primary source](#)

### S23

Kets, Pennock, Sethi and Shah (2014), Betting Strategies, Market Selection, and the Wisdom of Crowds. Primary AAAI paper. Wealth dynamics and belief aggregation are theory, not a live performance claim.

[Open primary source](#)

### S24

Dubach (2026), Anatomy of a Decentralized Prediction Market. Full arXiv preprint examined. Useful on book data and inferred trade-side limitations; no independent replication here.

[Open primary source](#)

# Repository and deck evidence ledger

## PINNED REVIEW SNAPSHOT; ONGOING WORK MAY HAVE ADVANCED

**No orders, credentials, remote runtime settings or frozen experiment files were changed for this review.**

### R1 - Grokbot operations

Repository: 17thgreen/Grokbot-Deathmatch-Dedicated-Repo. Main reviewed at 1825ec8c78e7f1444fa179ba9762558ece081144. Read its eight Markdown operations/handoff files and followed the explicit science-source link. No production process health or live account performance was independently checked.

[Open primary source](#)

### R2 - Canonical science main

Repository: 17thgreen/GPT-6-Astra-Deathmatch. Main a281adc944e4dacffcdb5677a140fabaed675a81. Reviewed AGENTS, README, experiment registry, Stern round-two results, NFL factorial and completion results, Q7/refiner status and policies, fee logic, and representative accepted-input requirements and empty scorecards in newer market-family labs. Not every duplicate fixture was independently rerun.

[Open primary source](#)

### R3 / R4 - Q7 and Q8 research branches

Q7: research/q7-paired-price at 01c726ae9244d801d359e00df67d6a71f4012669; nfl\_pair\_price\_lab\_20260921/Q7\_RESULTS.md and policy code. Q8: research/q8-joint-routing at d7079b6fece902e6e4603114a1d9ed5ef5a246c5; nfl\_joint\_route\_lab\_20260921/Q8\_RESULTS.md. Read saved results and compared relevant Q7 implementations; no controlled reconciliation run was performed.

### R5 - Cross-sport continuation

research/m2-continuation at f44574e8f318920cc8fdf87c9c3d35c18bca2107. Read cross\_sport\_replay\_20260922/M2\_RESULTS.md and market\_adaptation\_20260922/M3\_RESULTS.md. Small reused panels remain development evidence.

### R6 - All-market structure

research/all-market-structure-20260924 at a59c9e331dcfd71c957e753bc4fd2460df403c0. Read root results, recovery/next-study notes, threshold findings, triage/DECISIONS.md and depth\_survey/RESULTS.md. AMS004 observations span September 24, 03:32:10-03:35:44 UTC. Snapshot candidate rates do not measure persistence. Later census/tape scaffolds are not completed return studies.

### D1 - Supplied deck

Rajiv Sethi, Engines of Prophecy: The Power and Perils of Prediction Markets, D2 Columbia, May 2026. Uploaded 74-page PDF, Rajiv Sethi - Engines\_of\_Prophecy(1).pdf. All pages text-extracted and visually inspected in contact sheets. The supplied photo corresponds to page 32's two coauthored papers.